

ML Prac1
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Section: B_B2
Roll no: 22

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("/content/titanic (1).csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S	
3	4	1	1	Futrelle, Mrs. Jacques	female	35.0	1	3	1601	53.1000	C148	C	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.tail()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S	
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S	
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S	
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C	
890	891	0	3	Bonn, Mr. Louis	male	28.0	0	0	370829	7.75	NaN	C	

```
df.shape
```

(891, 12)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208	
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429	
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200	

df.isnull()

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
0	False	False	False	False	False	False	False	False	False	False	True	False	
1	False	False	False	False	False	False	False	False	False	False	False	False	
2	False	False	False	False	False	False	False	False	False	False	True	False	
3	False	False	False	False	False	False	False	False	False	False	False	False	
4	False	False	False	False	False	False	False	False	False	False	True	False	
...	...	...	...	...	...	...	...	...	...	...	...	...	
886	False	False	False	False	False	False	False	False	False	False	True	False	
887	False	False	False	False	False	False	False	False	False	False	False	False	
888	False	False	False	False	False	True	False	False	False	False	True	False	
889	False	False	False	False	False	False	False	False	False	False	False	False	
890	False	False	False	False	False	False	False	False	False	False	True	False	

891 rows × 12 columns

df.isnull().sum()

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
df["Embarked"] = df["Embarked"].replace("S", 0)
df["Embarked"] = df["Embarked"].replace("C", 1)
df["Embarked"] = df["Embarked"].replace("Q", 2)
```

df.head()

```
/tmp/ipykernel_2086/4092569315.py:3: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future release.
df["Embarked"] = df["Embarked"].replace("Q", 2)
```

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked		
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	0.0	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	1.0	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	0.0	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	0.0	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df["Sex"] = df["Sex"].replace("male", 0)
df["Sex"] = df["Sex"].replace("female", 1)

df.head()
```

```
/tmp/ipykernel_2086/140413152.py:2: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future release.
df["Sex"] = df["Sex"].replace("female", 1)
```

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
0	1	0	3	Braund, Mr. Owen Harris	0	22.0	1	0	A/5 21171	7.2500	NaN	0.0
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	1	38.0	1	0	PC 17599	71.2833	C85	1.0
2	3	1	3	Heikkinen, Miss. Laina	1	26.0	0	0	STON/O2. 3101282	7.9250	NaN	0.0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	1	35.0	1	0	113803	53.1000	C123	0.0

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df["Age"].fillna(df["Age"].median() , inplace=True)

df.head()
```

```
/tmp/ipykernel_2086/2278289005.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chain
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df
```

df["Age"].fillna(df["Age"].median() , inplace=True)

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
0	1	0	3	Braund, Mr. Owen Harris	0	22.0	1	0	A/5 21171	7.2500	NaN	0.0	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	1	38.0	1	0	PC 17599	71.2833	C85	1.0	
2	3	1	3	Heikkinen, Miss. Laina	1	26.0	0	0	STON/O2. 3101282	7.9250	NaN	0.0	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	1	35.0	1	0	113803	53.1000	C123	0.0	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df["Embarked"].fillna(df["Embarked"].mode()[0] , inplace=True)
```

```
/tmp/ipykernel_2086/3776449277.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chain
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df
```

```
df["Embarked"].fillna(df["Embarked"].mode()[0] , inplace=True)
```

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	0

dtype: int64

```
df.drop("Cabin",axis=1, inplace=True)

df.drop("PassengerId",axis=1, inplace=True)

df.drop("Name",axis=1, inplace=True)

df.drop("Ticket",axis=1, inplace=True)

df.head()
```

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	0	22.0	1	0	7.2500	0.0
1	1	1	1	38.0	1	0	71.2833	1.0
2	1	3	1	26.0	0	0	7.9250	0.0
3	1	1	1	35.0	1	0	53.1000	0.0
4	0	3	0	35.0	0	0	8.0500	0.0

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.duplicated().sum()
```

```
np.int64(116)
```

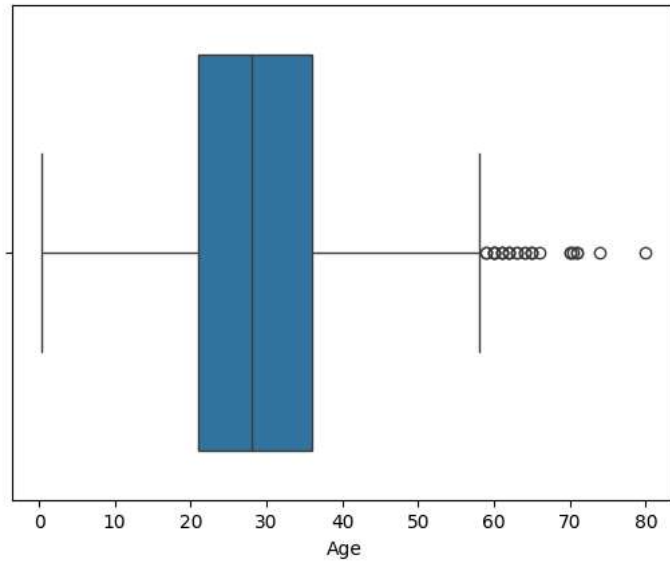
```
df.drop_duplicates(inplace=True)
```

```
sns.boxplot(x=df["Age"])
plt.show
```

```
matplotlib.pyplot.show
def show(*args, **kwargs) -> None
```

****Auto-show in jupyter notebooks****

The jupyter backends (activated via ``%matplotlib inline``, ``%matplotlib notebook``, or ``%matplotlib widget``), call ``show()`` at the end of every cell by default. Thus, you usually don't have to call it explicitly there.



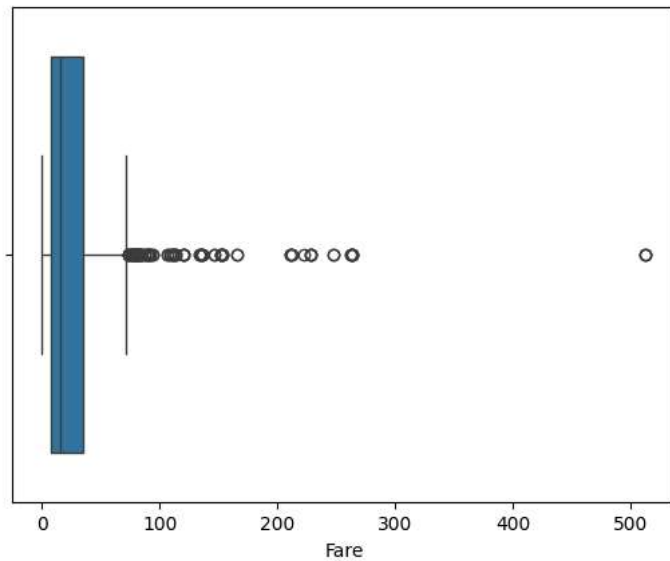
```
sns.boxplot(x=df["Fare"])
plt.show
```

```
matplotlib.pyplot.show
def show(*args, **kwargs) -> None
```

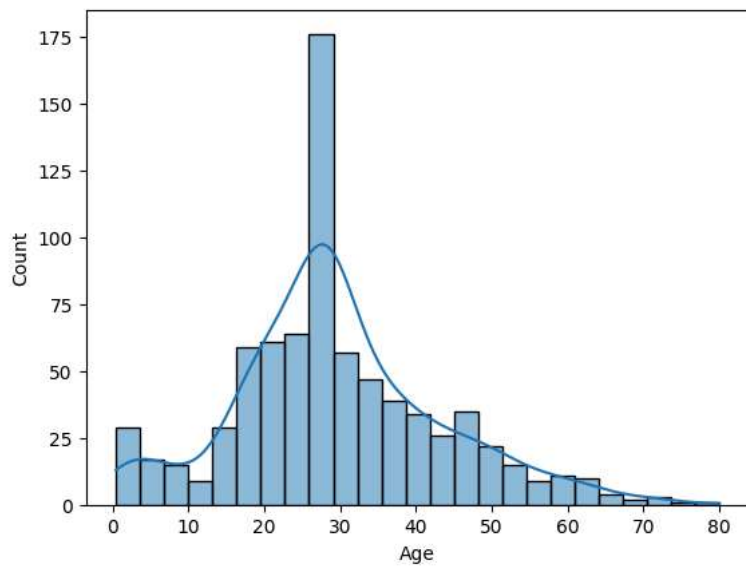
[/usr/local/lib/python3.12/dist-packages/matplotlib/pyplot.py](https://matplotlib.org/3.1.2/matplotlib.pyplot.py)
Display all open figures.

Parameters

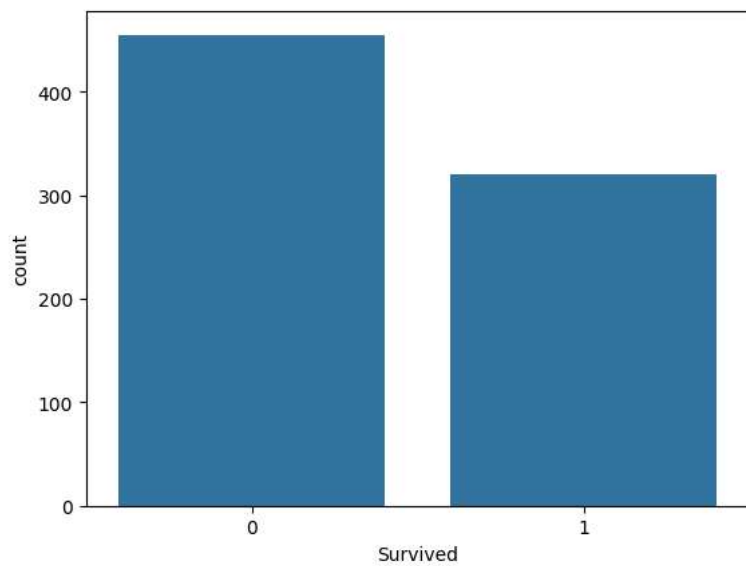
block : bool, optional



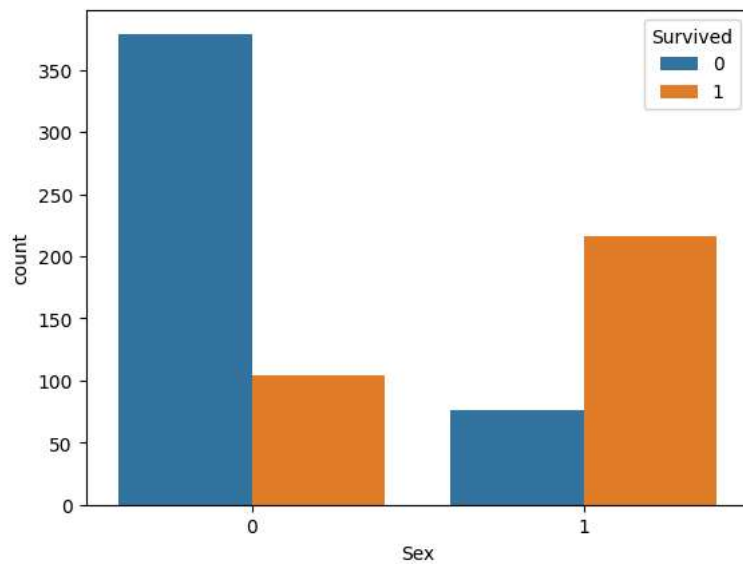
```
sns.histplot(df["Age"], kde = True)
plt.show()
```



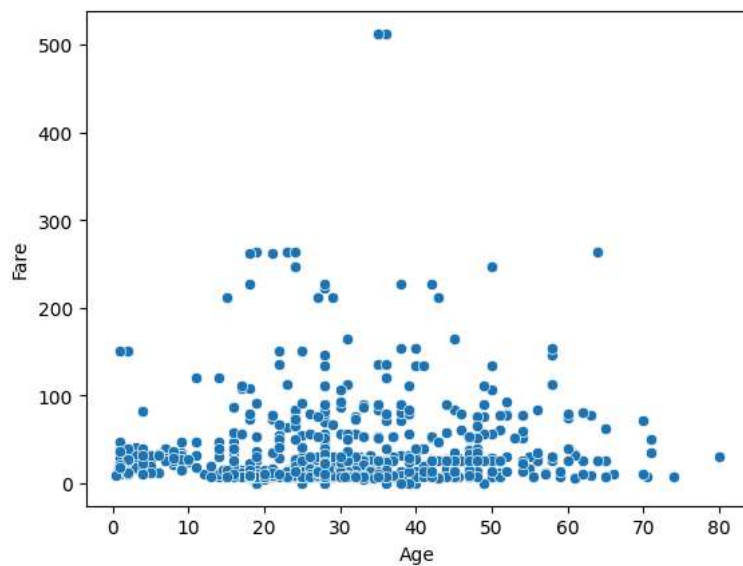
```
sns.countplot(x="Survived",data=df)  
plt.show()
```



```
sns.countplot(x="Sex",hue="Survived",data=df)  
plt.show()
```



```
sns.scatterplot(x="Age", y="Fare", data=df)
plt.show()
```

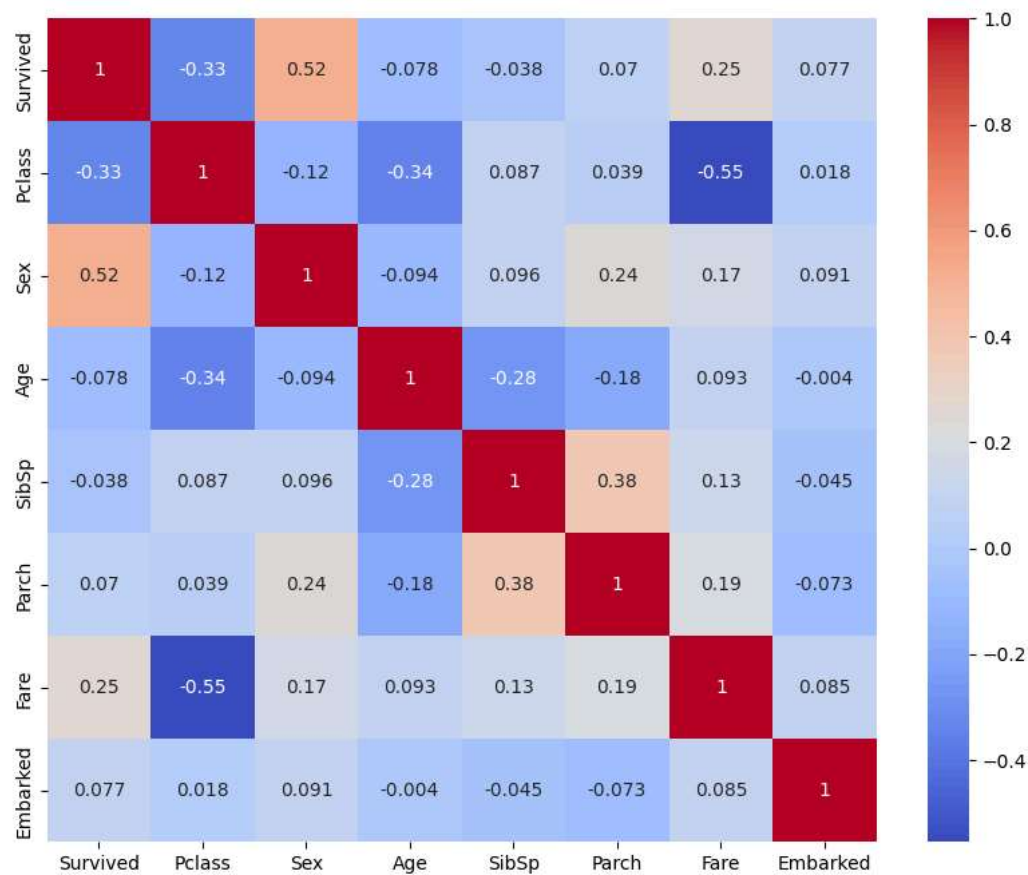


```
plt.figure(figsize=(10,8))

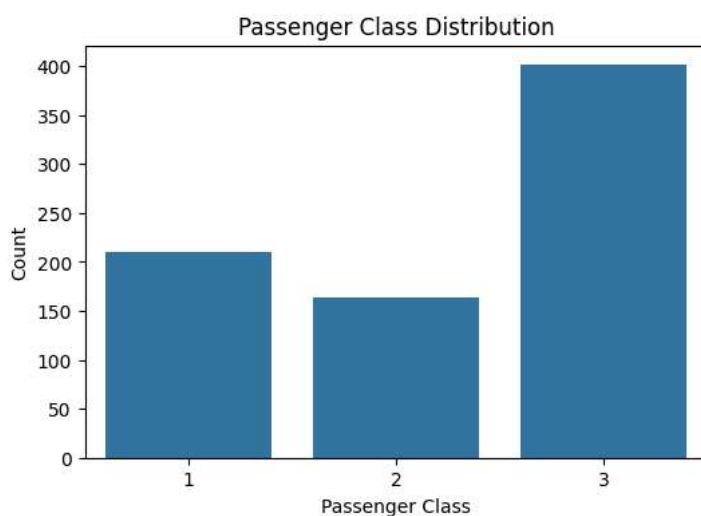
numeric_df = df.select_dtypes(include=['number'])

sns.heatmap(numeric_df.corr(),annot=True,cmap='coolwarm')

plt.show()
```

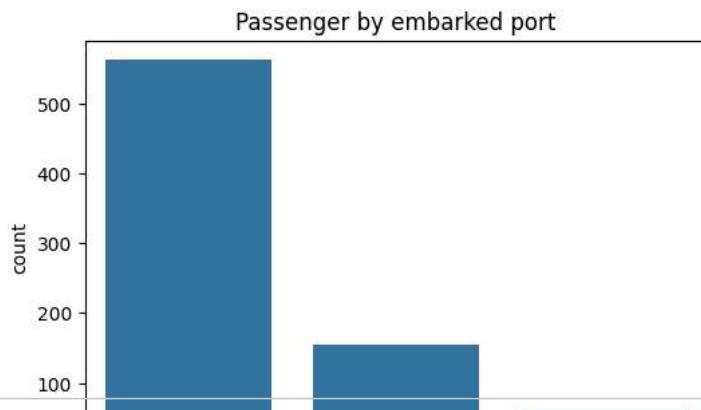


```
plt.figure(figsize=(6, 4))
sns.countplot(x="Pclass", data=df)
plt.title("Passenger Class Distribution")
plt.xlabel("Passenger Class")
plt.ylabel("Count")
plt.show()
```

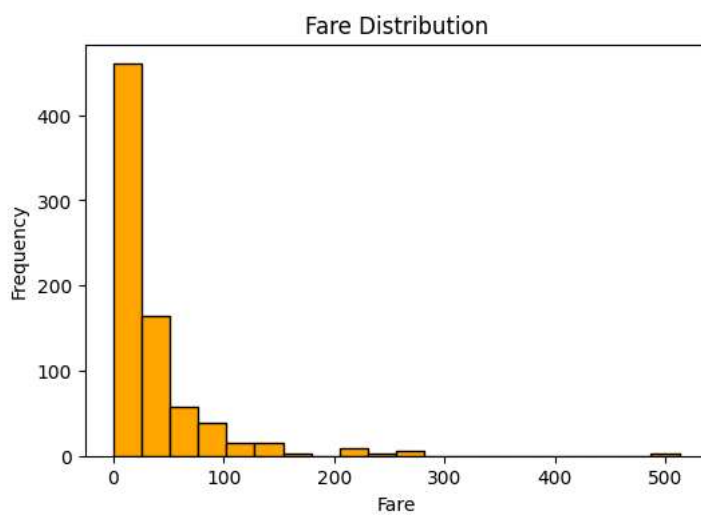


```
plt.figure(figsize=(6, 4))
sns.countplot(x="Embarked", data=df)
plt.title("Passenger by embarked port")

plt.show()
```



```
plt.figure(figsize=(6, 4))
plt.hist(df["Fare"], bins=20, color="orange", edgecolor="black")#histogram
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Frequency")
plt.show()
```



```
#feature scaling
X=df.drop("Survived",axis=1)
Y=df["Survived"]
```

```
from sklearn.preprocessing import StandardScaler

# Create scaler object
scaler = StandardScaler()

# Scale numerical columns
df[['Age', 'Fare']] = scaler.fit_transform(df[['Age', 'Fare']])

# Display first 15 rows
df.head(15)
```

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	0	-0.551060	1	0	-0.527515	0.0
1	1	1	1	0.611945	1	0	0.695086	1.0
2	1	3	1	-0.260308	0	0	-0.514627	0.0
3	1	1	1	0.393881	1	0	0.347909	0.0
4	0	3	0	0.393881	0	0	-0.512240	0.0
5	0	3	0	-0.114933	0	0	-0.504444	2.0
6	0	1	0	1.774040	0	0	0.324281	0.0